10A — Deep-Dive Audit: Economy & Supply Chain (Part 1 of 5: Primary Production, Food/Canteen & Bigu Lore, Mining, Herbology, Rare Organs & Reserve Intelligence)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'`)
- **Primary Audited Modules**: `js/economy.js`, `js/ai.js`, `js/sim.js`, `js/cpeco.js`, `js/content.js`, `js/sect.js`, `js/fishing.js`
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini**)
- **Core Design Invariant**: 100% Closed-Loop Emergent Economy (`Debit === Credit`, zero unbacked Spirit-Stone or material conjuration), authentic Xianxia progression context, and zero hard-locks on `js/human_core.js` (`155:2f9c9dd7`).

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1. Category 10 Master Roadmap (5-Part Economy & Trade Audit)

1. **`10A` (This Document — Primary Resource Supply Chain & Reserve Intelligence)**: Grain, Canteen, Specialty Meals, Immortal Inedia (`Bigu / 辟谷`), Mining, Herbology, Rare Beast Organs, and `inventoryIntelligence` / `shortageDirectives`.
2. **`10B` (Part 2 — Workshop Queues, Multi-Discipline Crafting & Recipe Lifecycle)**: `ECO.pickOrder`, `ALCHEMY / FORGING / TALISMAN / ARRAY` queue starvation, idempotent craft ledger `craftTransactions`, recipe authorization/copying/trading, and personal crafting.
3. **`10C` (Part 3 — Contribution Point `CP` Monetary Engine `cpeco.js`)**: Minting vs. burning velocity, `c.price` inflation/deflation dynamics, `stone_grant` exchange rate, welfare favoritism, and wealth inequality across ranks.
4. **`10D` (Part 4 — Visiting Merchants & External Trade `external_market.js`)**: Sect `cashReserve` buffering, `buyList` vs. surplus liquidation, member-to-merchant Spirit-Stone trading, and inter-sect treaty price multipliers.
5. **`10E` (Part 5 — Grand Finale: Master Closed-Loop Economic Synthesis)**: Complete Spirit-Stone conservation ledger across all 40 scripts, `CP <-> Stone` arbitrage checks, 200-year macro equilibrium, and prioritized `P0 -> P3` engineering roadmap.

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2. [ECO-FOOD-01] Five Logic & Xianxia-Context Defects in Food, Canteen, Specialty Meals & Fasting (Bigu)

2.1 Zero Xianxia Inedia (Bigu / 辟谷) Scaling: 2,000-Year Stage-9 Immortals Eat 3 Bowls of Porridge Daily & Gridlock the Canteen

- **Code Trace (`js/ai.js` — `AI.needs`)**:
  - Satiety decays at a uniform flat rate per tick for every living NPC regardless of cultivation realm (`Stage 1` Qi Refining through `Stage 9` Immortal Ascension).
- **Xianxia Lore Mismatch & 80-NPC Chain-Reaction**:
  - In classical Xianxia lore, high-realm cultivators (`Golden Core / Nascent Soul / Tribulation Immortal`) practice **`Bigu`** **(辟谷 — Fasting on Heaven-and-Earth Qi)** because mundane mortal grain (五谷) carries earthly turbidity, relying instead on ambient Qi, Spirit Tea, Spirit-Fish Banquets, or Fasting Elixirs.
  - Mechanically, in a mature 80-NPC sect where `canteen` only has `16 eat` seats (`world_map.js:102`), 30 high-realm elders walking down from distant mountains 3 times a day clog Canteen seats and steal mundane meals from Rank-0 mortal farmers and miners—pushing laborers below `satiety < 25` and triggering farm productivity collapse.
- **Remediation Blueprint**:
  - Scale base satiety decay smoothly by `CULT.equivalentStage(n)` when `n.h.state.qi > CULT.qiMax(n) * 0.25` (excluding `BODY` / `DEVOURING_FLESH` / `ARENA_GLORY` physical furnace paths, which genuinely require high caloric/spirit-meat intake!):
    - Mortal / Stage 1–2: `1.00×`
    - Stage 3–4: `0.75×`
    - Stage 5–6: `0.45×`
    - Stage 7–9 (`Bigu` Mastery): `0.20×` (sustained primarily by ambient Qi).

2.2 Expeditions & Outpost Convoys Wipe Out Gourmet `mealSpecialties` Into Plain Dry Rations (`js/economy.js`)

- **Code Trace**:
  - When a `SPIRIT_FEAST_HEALER` or `JADE_MEAL_PREPARER` chef cooks, `ECO.finishCooking` pushes `{ role, servings, potency, until }` into `W.sect.mealSpecialties`.
  - However, `ECO.take(W, 'meal', n)` decrements `W.sect.mealSpecialties` whenever **any** caller takes `'meal'`—including bulk withdrawals by `EXPED.depart` (`15–30` travel rations) and `SECT.OUTPOSTS_LOGISTICS.dispatchSupplyConvoy`!

    - Neither `EXPED.travelStep` nor `tickOutpostLogistics` ever applies `RECOVERY_FOOD` / `CULTIVATION_FOOD` buffs! Thus, a single departing caravan silently destroys 24 servings of freshly cooked gourmet banquet buffs without anyone receiving a single stat point.
- **Remediation Blueprint**:
    - Only consume `W.sect.mealSpecialties` inside `ECO.reserveMeal` / `ECO.finishMeal` (actual Canteen/Pantry dining), or attach `hasSpecialtyBuff` to expedition rations so field parties actually receive the chef's buff!
    - Also tag the `professionMeal` buff with `kind: 'meal'` so `ST.preFight` (`stats.js:663`) only blocks duplicate `kind === 'cbPill'` buffs instead of locking fed disciples out of all combat pills (`see 06D`).

2.3 Famine Panic Blindness to Spirit-Fish Cuisine & Winter Granary Cliff (`js/sim.js:518`, `js/ai.js`, `js/economy.js`)

- **Root Cause**:
    1. `SIM.dailyUpdate` (`sim.js:518`) triggers emergency panic grain purchases (`ECO.pay` without crediting `W.world.resourcePool.stones`!) when `meal` and `grain` hit `0`, ignoring cooked Spirit-Fish dishes (`FISHING.totalCookedDishCount(W)`).
    2. `W.sect.reserveDirectives.minGrainDays` (`14` days) is season-blind. Because `Season 3` (Winter) slashes `A.farm` yield to `0.35×`, failing to raise the Autumn (`Season 2`) reserve target to `22` days causes recurring late-Winter food panics in 60–80 NPC sects.

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3. `[ECO-HARV-02]` Primary Harvesting Defects: Infinite Unbacked Mine Stones, Tier-3 Facility Cap & Pool Contamination

3.1 Dual Unbacked Spirit-Stone Printers in `A.mine` (`js/economy.js` & `js/ai.js:592`)

- **Defect 1 (`ECO.mineYield`)**: Credits mined Spirit Stones via `ECO.earn(W, amt, ...)` without deducting from `W.world.resourcePool.stones` (unlike Contestable Resource Nodes and Micro-Locations). Over a 100-year run, sect mining continuously inflates global Spirit-Stone supply without bound.
- **Defect 2 (`A.mine.done`, `ai.js:592`)**:

```javascript
if (n.cult && n.cult.track && CULT.checkTrackGates && !CULT.checkTrackGates(W,
n).resourceOk) {
  n.inv.stones = (n.inv.stones || 0) + Math.min(18, Math.max(5, (ctx.got || 1) *
6));
}
```

Even when `ctx.got === 0` (`0` ores mined!), `(0 || 1) * 6` conjures `6` free personal Spirit Stones out of thin air exclusively for disciples with an unmet track gate, while normal miners get `0` personal stones.

- **Remediation Blueprint**:
  - Deduct all mined Spirit Stones (`both sect share and a fair 10%–15% miner personal finder's share when ctx.got > 0`) strictly from `W.world.resourcePool.stones` (with bounded geological vein regeneration in `08F`), and remove the `ctx.got || 1` conjuration hack.

3.2 Hard Tier-3 Ceiling on Upgraded Lv.4–5 Sect Mines & Herb Gardens (`js/economy.js:685–718`)

- Upgrading `herb_garden` or `mine` to `Lv.4–5` increases station count/speed, but `ECO.harvestHerb` and `ECO.mineYield` never roll Tier-4 materials (`max tier = 3`), creating a sharp cliff between T3 domestic production and T4 procedural/alchemy recipes.
- **Fix**: When `building.level >= 4` and worker skill `>= 65`, allow a `4%–10%` rare proc for Tier-4 regional herbs/ores (bounded by daily vein richness).

3.3 Non-Botanical / Non-Mineral Pool Contamination (`economy.js:354–355`, `cpeco.js:138`)

- As cross-referenced in `04C` & `05G`, `MAT_BRUSH_SPIRIT_WEASEL_HAIR_1` inside `ORES[1]` and `MAT_WHIP_SHADOW_PYTHON_TENDON_3` / `IRON_SILKWORM_THREAD_1` / `ZITHER_PARASOL_WOOD_1` inside `HERBS` & `HERB_POOL` dilute medicinal herb output by `35%–60%`. Relocate animal pelts/tendons to `WILD` / `SPIRIT_HUSBANDRY` and timber to `grove` / `WILD_RES WOOD` nodes.

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4. [ECO-ORG-03] Shadow Inventory Isolation of Rare Beast Organs (W.sect.organs & n.inv.organs in js/content.js vs js/cpeco.js)

- **Root Cause**: content.js stores rare beast organs in W.sect.organs and n.inv.organs rather than W.sect.stock / n.inv.mats, and three core economic loops forgot to bridge that separate namespace:
    1. **CPE.donorValue (cpeco.js:76–85)**: Sums out.mats, out.pills, out.gear, out.techs, and out.stones on expedition handover—**completely omitting out.organs**, so hunting parties earn **0 CP** for dissecting mutated beast organs!
    2. **A.donate (cpeco.js:176–260)**: Checks personalMats, personalTechs, personalRecipes, and personalSurplusGear—**completely omitting n.inv.organs**, so organs placed in personal bags are never donated via the standard donation act (CONTENT.donateOrgan is never called by A.donate)!
    3. **ECO.inventoryIntelligence**: Does not include W.sect.organs in shortage/surplus reports.
- **Remediation Blueprint**:
    - Add out.organs valuation (10 * organ.tier * qty) into CPE.donorValue, include n.inv.organs in A.donate.can / score / done (delegating to CONTENT.donateOrgan), and surface W.sect.organs in ECO.inventoryIntelligence.

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5. [ECO-INTEL-04] inventoryIntelligence & shortageDirectives Blind Spots and Sequence Desyncs

2. **Blindness to Specialization Track Material Bottlenecks**:
    - SECT.hallHeads populates W.sect.shortageDirectives and W.sect.buyList from basic provisions, emergency pills, and active hall orders, **but ignores materials needed by core disciples stuck at CULT.checkTrackGates(W, n)** (e.g., missingMats) and **ignores experiment.trialInputs (content.js)**. Adding aggregated member track-gate material deficits (priority: 'normal') lets Visiting Merchants (EXT.openVisit & EXT.tradeSect) organically supply bottlenecked disciples!
3. **Direct W.sect.stock Writes Bypassing W._stockSeq / _pillSeq / _stockDirty**:
    - Direct mutations (e.g., W.sect.stock[k]++ in combat.js:ANATOMY_BREAKER and sim.js:retire) fail to increment

`W._stockSeq` and `W._pillSeq`, leaving `W._pillStockKeys` (`cpeco.js:264`), `W._pbMask` (`cpeco.js:112`), and `W.sect.inventoryIntelligence` stale. Enforce routing 100% of stock mutations through `ECO.add` / `ECO.take`.